14.1: Mendel used the scientific approach to identify two laws of inheritance

Modern genetics began during the mid-1800s with a monk named Gregor Mendel, who discovered the basic principles of heredity by breeding garden peas in carefully planned experiments. As we retrace his work, you will recognize the key elements of the scientific process that were introduced in Concept 1.3.

Mendel's Experimental, Quantitative Approach

Mendel grew up on his parents' small farm in a region of Austria that is now part of the Czech Republic. In this agricultural area, Mendel received agricultural training in school along with a basic education. As an adolescent, Mendel overcame financial hardship and illness to excel in high school and, later, at the Olmutz Philosophical Institute.

In 1843, at the age of 21, Mendel entered an Augustinian monastery, a reasonable choice at that time for someone who valued the life of the mind. He considered becoming a teacher but failed the necessary examination. In 1851, he left the monastery to pursue two years of study in physics and chemistry at the University of Vienna. These were very important years for Mendel's development as a scientist, in large part due to the strong influence of two professors. One was the physicist Christian Doppler, who encouraged his students to learn science through experimentation and trained Mendel to use mathematics to help explain natural phenomena. The other was a botanist named Franz Unger, who aroused Mendel's interest in the causes of variation in plants.

After attending university, Mendel returned to the monastery and was assigned to teach at a local school, where several other instructors were enthusiastic about scientific research. In addition, his fellow monks shared a long-standing fascination with the breeding of plants. Around 1857, Mendel began breeding garden peas in the abbey garden to study inheritance. Although the question of heredity had long been a focus of curiosity at the monastery, Mendel's fresh approach allowed him to deduce principles that had remained elusive to others.

One reason Mendel probably chose to work with peas is that there are many varieties. For example, one variety has purple flowers, while another variety has white flowers. A heritable feature that varies among individuals, such as flower color, is called a **character**. Each variant for a character, such as purple or white color for flowers, is called a **trait**.

Other advantages of using peas are their short generation time and the large number of offspring from each mating. Furthermore, Mendel could strictly control mating between plants **(Figure 14.2)**. Each pea flower

has both pollen-producing organs (stamens) and an egg-bearing organ (carpel). In nature, pea plants usually self-fertilize: Pollen grains from the stamens land on the carpel of the same flower, and sperm released from the pollen grains fertilize eggs present in the carpel.* To achieve cross-pollination of two plants, Mendel removed the immature stamens of a plant before they produced pollen and then dusted pollen from another plant onto the altered flowers (see Figure 14.2). Each resulting zygote then developed into a plant embryo encased in a seed (a pea). His method allowed Mendel to always be sure of the parentage of new seeds.

Figure 14.2

Research Method: Crossing Pea Plants

Application

By crossing, or mating, two true-breeding varieties of an organism, scientists can study patterns of inheritance. In this example, Mendel crossed pea plants that varied in flower color.

Technique

A diagram of crossing pea plants.

One. Removed stamens from a purple flower.

Two. Transferred sperm-bearing pollen from the stamens of a white flower to the egg-bearing carpel of a purple flower.

Three. Waited for pollinated carpel to mature into the pod.

Four. Planted seeds from pod.

Results

When pollen from a white flower was transferred to a purple flower, the first-generation hybrids all had purple flowers. The result was the same for the reciprocal cross, which involved the transfer of pollen from purple flowers to white flowers.

The results of the pea plant experiment. Five. First filial generation offspring, F 1. Examined offspring: all had purple flowers.

Mendel chose to track only those characters that occurred in two distinct, alternative forms, such as purple or white flower color. He also made sure that he started his experiments with varieties that were **true-**

breeding—that is, over many generations of self-pollination, these plants had produced only the same variety as the parent plant. For example, a plant with purple flowers is true-breeding if the seeds produced by self-pollination in successive generations all give rise to plants that also have purple flowers.

In a typical breeding experiment, Mendel cross-pollinated two contrasting, true-breeding pea varieties—for example, purple-flowered plants and white-flowered plants (see Figure 14.2). This mating, or *crossing*, of two true-breeding varieties is called **hybridization**. The true-breeding parents are referred to as the **P generation** ****(parental generation), and their hybrid offspring are the ****F1 generation** (first filial generation, the word *filial* from the Latin word for "son"). Allowing these F1 hybrids to self-pollinate (or to cross-pollinate with other F1 hybrids) produces an **F2 generation** (second filial generation). Mendel usually followed traits for at least the P, F1, and F2 generations. Had he stopped his experiments with the F1 generation, the basic patterns of inheritance would have eluded him. Mendel's quantitative analysis of the F2 plants from thousands of genetic crosses like these allowed him to deduce two fundamental principles of heredity, now called the law of segregation and the law of independent assortment.

The Law of Segregation

The explanation of heredity most widely in favor during the 1800s was the "blending" hypothesis, the idea that genetic material contributed by the two parents mixes just as blue and yellow paints blend to make green. This hypothesis predicts that over many generations, a freely mating population will give rise to a uniform population of individuals, something we don't see.

If the blending model were correct, the F1 hybrids from Mendel's cross between purple-flowered and white-flowered pea plants would have been pale purple flowers, a trait intermediate between those of the P generation. Notice in Figure 14.2 that the experiment produced a very different result: All the F1 offspring had flowers of the same color as the purple-flowered parents. What happened to the white-flowered plants' genetic contribution to the hybrids? If it were lost, then the F1 plants could produce only purple-flowered offspring in the F2 generation. But when Mendel allowed the F1 plants to self- or cross-pollinate and planted their seeds, the white-flower trait reappeared in the F2 generation. The blending hypothesis is also inconsistent with this reappearance of traits after they've skipped a generation.

Mendel used very large sample sizes and kept accurate records of his results: 705 of the F2 plants had purple flowers, and 224 had white flowers. These data fit a ratio of approximately three purple to one white **(Figure 14.3)**. Mendel reasoned that the heritable factor for white flowers did not disappear in the F1 plants but was somehow hidden, or masked, when the purple-flower factor was present. In Mendel's terminology, purple flower color is a *dominant* trait, and white flower color is a *recessive* trait. The reappearance of white-flowered plants in the F2 generation was evidence that the heritable factor causing white flowers had not been diluted or destroyed by coexisting with the purple-flower factor in the F1 hybrids.

Mendel observed the same pattern of inheritance in six other characters, each represented by two distinctly different traits **(Table 14.1)**. For example, when Mendel crossed a true-breeding variety that produced smooth, round pea seeds with one that produced wrinkled seeds, all the F1 hybrids produced round seeds; this is the dominant trait for seed shape. In the F2 generation, approximately 75% of the seeds were round and 25% were wrinkled—a 3:1 ratio, as in Figure 14.3. Now let's see how Mendel deduced the law of segregation from his experimental results. In the discussion that follows, we will use modern terms instead of some of the terms used by Mendel. (For example, we'll use "gene" instead of Mendel's "heritable factor.")

Figure 14.3

Inquiry: When F1 hybrid pea plants self- or cross-pollinate, which traits appear in the F2 generation?

Experiment

Mendel crossed true-breeding purple-flowered plants and white-flowered plants (crosses are symbolized by ×). The resulting F1 hybrids were allowed to self-pollinate or were cross-pollinated with other F1 hybrids. The F2 generation plants were then observed for flower color.

P generation (true-breeding parents): purple flowers, crossed with, white flowers. F 1 generation (hybrids): all plants had purple flowers; self- or cross-pollination. F 2 generation: 705 purple-flowered plants, 224 white-flowered plants.

Results

Both purple-flowered and white-flowered plants appeared in the F2 generation, in a ratio of approximately 3:1.

Conclusion

The "heritable factor" for the recessive trait (white flowers) had not been destroyed, deleted, or "blended" in the F1 generation but was merely masked by the presence of the factor for purple flowers, which is the dominant trait.

Data from G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brünn* 4:3–47 (1866).

WHAT IF? *If you mated two purple-flowered plants from the P generation, what ratio of traits would you expect to observe in the offspring? Explain. What might Mendel have concluded if he stopped his experiment after the* F_1 *generation?*

*For suggested answer, see*Appendix A.*

Table 14.1

The Results of Mendel's F1 Crosses for Seven Characters in Pea Plants

Table 14.1 gives the results of Mendel's F 1 crosses for seven characters in pea plants.

The table is as follows:

Character	Dominant trait	Recessive trait	F2 generation dominant: recessive	Ratio
flower color	purple	white	705 to 224	3.15 to 1
seed color	yellow	green	6,022 to 2,001	3.01 to 1
seed shape	round	wrinkled	5,474 to 1,850	2.96 to 1
pod color	green	yellow	428 to 152	2.82 to 1
pod shape	inflated	constricted	882 to 299	2.95 to 1
flower position	axial	terminal	651 to 207	3.14 to 1
stem length	tall	dwarf	787 to 277	2.84 to 1

Mendel's Model

Mendel developed a model to explain the 3:1 inheritance pattern that he consistently observed among the F2 offspring in his pea experiments. We'll now explore four related concepts making up Mendel's model, the fourth of which is the law of segregation.

First, *alternative versions of genes account for variations in inherited characters*. The gene for flower color in pea plants, for example, exists in two versions, one for purple flowers and the other for white flowers. These alternative versions of a gene are called **alleles**. Today, we can relate this concept to chromosomes

and DNA. As shown in **Figure 14.4**, each gene is a sequence of nucleotides at a specific place, or locus, along a particular chromosome. The DNA at that locus, however, can vary slightly in its nucleotide sequence. This variation in information content can affect the function of the encoded protein and thus an inherited character of the organism. The purple-flower allele and the white-flower allele are two DNA sequence variations possible at the flower-color locus on a pea plant's chromosomes. The purple-flower allele sequence allows synthesis of purple pigment, and the white-flower allele sequence does not.

Figure 14.4

Alleles, alternative versions of a gene.

Homologous chromosomes with gene for flower color; arrows zoom in on D N A sequences for purple and white flower alleles. A functional enzyme is made from the purple flower allele only.

A pair of homologous chromosomes, one paternal and one maternal, with a region on each chromosome marked: an allele for purple flower color on the paternal chromosome and the allele for white flowers on the maternal chromosome. Arrows from each allele zoom in on the D N A molecules present on each chromosome at this location; the nucleotide sequence shown in the paternal (purple flower) allele is C T A A A T C G G T; the nucleotide sequence in the maternal (white flower) allele is A T A A A T C G G T. The D N A sequence in the paternal (purple flower) allele results, through a series of steps, in the production of an enzyme that helps synthesize purple pigment. The D N A sequence in the maternal (white flower) allele results in the absence of the enzyme. The interaction between these two alleles results in purple flower color because one purple flower allele results in sufficient pigment to produce purple flowers.

This diagram shows a pair of homologous chromosomes in an F1 hybrid pea plant, with the actual DNA sequence from the flower-color allele of each chromosome. The paternally inherited chromosome (blue) has an allele for purple flowers, which codes for a protein that indirectly controls synthesis of purple pigment. The maternally inherited chromosome (red) has an allele for white flowers, which results in no functional protein being made.

Figure 14.4 Walkthrough

Alleles, alternative versions of a gene

Watch an explanation of this figure in the Figure 14.4 Walkthrough: Alleles, alternative versions of a gene.

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Second, *for each character, an organism inherits two versions (that is, two alleles) of a gene, one from each parent*. Remarkably, Mendel made this deduction without knowing about the role, or even the existence, of chromosomes. Each somatic cell in a diploid organism has two sets of chromosomes, one set inherited from each parent (see Figure 13.4). Thus, a genetic locus is actually represented twice in a diploid cell, once on each homolog of a specific pair of chromosomes. The two alleles at a particular locus may be identical, as in the true-breeding plants of Mendel's P generation. Or the alleles may differ, as in the F1 hybrids (see Figure 14.4).

Third, *if the two alleles at a locus differ, then one, the* ***dominant allele***, *determines the organism's appearance; the other, the* ***recessive allele***, *has no noticeable effect on the organism's appearance*. Accordingly, Mendel's F1 plants had purple flowers because the allele for that trait is dominant and the allele for white flowers is recessive.

The fourth and final part of Mendel's model, the **law of segregation**, states that *the two alleles for a heritable character segregate (in other words, separate from each other) during gamete formation and end up in different gametes*. Thus, an egg or a sperm gets only one of the two alleles that are present in the diploid cells of the organism making the gamete. In terms of chromosomes, this segregation corresponds to the distribution of copies of the two members of a pair of homologous chromosomes to different gametes in meiosis (see Figure 13.7). Note that if an organism has identical alleles for a particular character, then that allele is present in all gametes. Because it is the only allele that can be passed on to offspring when the plant self-pollinates, the offspring always look the same as their parents in regard to that characteristic; this explains why these plants are true-breeding. But if different alleles are present, as in the F1 hybrids, then 50% of the gametes receive the dominant allele and 50% receive the recessive allele.

Does Mendel's segregation model account for the 3:1 ratio he observed in the F2 generation of his numerous crosses? For the flower-color character, the model predicts that the two different alleles present in an F1 individual will segregate into gametes such that half the gametes will have the purple-flower allele and half will have the white-flower allele. During self-pollination, gametes of each class unite randomly. An egg with a purple-flower allele has an equal chance of being fertilized by a sperm with a purple-flower allele or by a sperm with a white-flower allele. Since the same is true for an egg with a white-flower allele, there are four equally likely combinations of sperm and egg. **Figure 14.5** illustrates these combinations using a **Punnett square**, a handy diagrammatic device for predicting the allele composition of offspring from a cross between individuals of known genetic makeup. Notice that we use a capital letter to symbolize a dominant allele and a lowercase letter for a recessive allele. In our example, P is the purple-flower allele, and p is the white-flower allele; it is often useful as well to be able to refer to the gene itself as the P/p gene.

Figure 14.5

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Mendel’s law of segregation.
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Mendel’s law of segregation with 3 generations.

Each generation is as follows.

- P Generation. Each true-breeding plant of the parental generation has two identical alleles, denoted as either PP both uppercase (purple) or pp both lowercase (white). Purple and white flowers are crossed. Genetic makeup: PP both uppercase (purple) and pp both lowercase (white). Gametes (shown as circles) each contain only one allele for the flower-color gene. In this case, every gamete produced by a given parent has the same allele. Gametes: uppercase P for purple (both uppercase PP) flowers and lowercase p for white (pp both lowercase) flowers.
- F 1 Generation. All flowers appear purple. Genetic makeup: P p (uppercase P lowercase P). Gametes: one-half uppercase P and one-half lowercase p. Union of parental gametes produces F 1 hybrids having an uppercase P lowercase p combination. Because the purple-flower allele is dominant, all these hybrids have purple flowers. When the hybrid plants produce gametes, the two alleles segregate. Half of the gametes receive the uppercase P allele and the other half the lowercase p allele.
- F 2 Generation. Punnett square showing possible F 2 genotypes. Outside square are gametes from F 1 parents. Result is purple to white flowers in 3 to 1 ratio in the F 2 generation. This box, a Punnett square, shows all possible combinations of alleles in offspring that result from an F 1 by F 1 (uppercase P lowercase p by uppercase P lowercase p) cross. Each square represents an equally probable product of fertilization. For example, the bottom left box shows the genetic combination resulting from a lowercase p egg fertilized by a sperm. Random combination of the gametes results in the 3 to 1 ratio that Mendel observed in the F2 generation.

This diagram shows the genetic makeup of the generations in Figure 14.3. It illustrates Mendel’s model for inheritance of the alleles of a single gene. Each plant has two alleles for the gene controlling flower color, one allele inherited from each of the plant’s parents. To construct a Punnett square that predicts the F2 generation offspring, we list all the possible gametes from one parent (here, the F1 female) along the left side of the square and all the possible gametes from the other parent (here, the F1 male) along the top. The boxes represent the offspring resulting from all the possible unions of male and female gametes.

Animation

Mendel's Cross of One Character: Flower Color

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Simplified Crosses of One Character in Humans

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Animation

Cross of One Character in “MendAliens”

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In the F2 offspring, what color will the flowers be? One-fourth of the plants have inherited two purple-flower alleles; clearly, these plants will have purple flowers. One-half of the F2 offspring have inherited one purple-flower allele and one white-flower allele; these plants will also have purple flowers, the dominant trait. Finally, one-fourth of the F2 plants have inherited two white-flower alleles and will express the recessive trait. Thus, Mendel’s model accounts for the 3:1 ratio of traits that he observed in the F2 generation.

Useful Genetic Vocabulary

An organism that has a pair of identical alleles for a gene encoding a character is called a **homozygote** and is said to be **homozygous** for that gene. In the parental generation in Figure 14.5, the purple-flowered pea plant is homozygous for the dominant allele (PP), while the white plant is homozygous for the recessive allele (pp). Homozygous plants “breed true” because all of their gametes contain the same allele—either P or p in this example. If we cross dominant homozygotes with recessive homozygotes, every offspring will have two different alleles—Pp in the case of the F1 hybrids of our flower-color experiment (see Figure 14.5). An organism that has two different alleles for a gene is called a **heterozygote** and is said to be **heterozygous**

for that gene. Unlike homozygotes, heterozygotes produce gametes with different alleles, so they are not true-breeding. For example, P- and p-containing gametes are both produced by our F1 hybrids. Self-pollination of the F1 hybrids thus produces both purple-flowered and white-flowered offspring.

Because of the different effects of dominant and recessive alleles, an organism's traits do not always reveal its genetic composition. Therefore, we distinguish between an organism's appearance or observable traits, called its **phenotype**, and its genetic makeup, its **genotype**. As shown in Figure 14.5 for the case of flower color in pea plants, PP and Pp plants have the same phenotype (purple flowers) but different genotypes. **Figure 14.6** reviews these terms. Note that the term *phenotype* refers not only to traits that relate directly to physical appearance but also to physiological traits. For example, one pea variety lacks the normal ability to self-pollinate, which is a phenotypic trait (called non-self-pollination).

Figure 14.6

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Phenotype versus genotype.
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Phenotypes are purple or white. Two different possible genotypes for purple flowers (PP both uppercase or uppercase P lowercase p) and one for white flowers (pp both lowercase).

Phenotype: In the F 2 offspring, there is a 3 to 1 ratio of purple to white flowers.

Genotype: in the F 2 offspring, there is a 1 to 2 to 1 ratio: 1 P P both uppercase (homozygous) to 2 uppercase P lowercase p (heterozygous) to 1 p p both lowercase (homozygous). The P P both uppercase and uppercase P lowercase p genotypes both result in purple flower phenotypes while the p p both lowercase genotype results in the white flower phenotype.

Grouping F2 offspring from a cross for flower color according to phenotype results in the typical 3:1 phenotypic ratio. In terms of genotype, however, there are actually two categories of purple-flowered plants, PP (homozygous) and PP (heterozygous), giving a 1:2:1 genotypic ratio.

The Testcross

Given a purple-flowered pea plant, we cannot tell if it is homozygous PP or heterozygous Pp because both genotypes result in the same purple phenotype. To determine the genotype, we can cross this plant with a white-flowered plant pp, which will make only gametes with the recessive allele p. The allele in the gamete contributed by the purple-flowered plant of unknown genotype will therefore determine the appearance of the offspring **(Figure 14.7)**. If all the offspring of the cross have purple flowers, then the purple-flowered mystery plant must be homozygous for the dominant allele, because a PP×pp cross produces all Pp offspring. But if both the purple and the white phenotypes appear among the offspring, then the purple-flowered

parent must be heterozygous. The offspring of a Pp×pp cross will be expected to have a 1:1 phenotypic ratio. Breeding an organism of unknown genotype with a recessive homozygote is called a **testcross** because it can reveal the genotype of that organism. The testcross was devised by Mendel and continues to be used by geneticists.

Figure 14.7

Research Method: The Testcross

Application

An organism that shows a dominant trait in its phenotype, such as purple flowers in pea plants, can be either homozygous for the dominant allele or heterozygous. To determine the organism's genotype, geneticists can perform a testcross.

Technique

In a testcross, the individual with the unknown genotype is crossed with a homozygous individual expressing the recessive trait (white flowers in this example), and Punnett squares are used to predict the possible outcomes.

A diagram of the experiment.

A purple flower is crossed with a white flower. The purple flower has dominant phenotype, unknown phenotype, uppercase P uppercase P or uppercase P lowercase p? White flower has recessive phenotype, known genotype, lower p lower p. Two Punnett squares are used to make predictions. First, if the purple flower parent is uppercase P lowercase P, the prediction is all purple flowers. If purple-flowered parent is uppercase P lowercase p, the prediction is half purple flowers and half white flowers.

Results

Matching the results to either prediction identifies the unknown parental genotype (either PP or Pp in this example). In this testcross, we transferred pollen from a white-flowered plant to the carpels of a purple-flowered plant; the opposite (reciprocal) cross would have led to the same results.

All offspring purple or half offspring purple and half offspring white.

Animation

Testcross in “MendAliens”

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The Law of Independent Assortment

Mendel derived the law of segregation from experiments in which he followed only a *single* character, such as flower color. All the F1 progeny produced in his crosses of true-breeding parents were **monohybrids**, meaning that they were heterozygous for the one particular character being followed in the cross. We refer to a cross between such heterozygotes as a **monohybrid cross**.

Mendel worked out the second law of inheritance by following *two* characters at the same time, such as seed color and seed shape. Seeds (peas) may be either yellow or green. They also may be either round (smooth) or wrinkled. From single-character crosses, Mendel knew that the allele for yellow seeds is dominant (Y), and the allele for green seeds is recessive (y). For the seed-shape character, the allele for round is dominant (R), and the allele for wrinkled is recessive (r).

Imagine crossing two true-breeding pea varieties that differ in *both* of these characters—a cross between a plant with yellow round seeds (YYRR) and a plant with green wrinkled seeds (yyrr). The F1 plants will be **dihybrids**, individuals heterozygous for the two characters being followed in the cross (YyRr). But are these two characters transmitted from parents to offspring as a package? That is, will the Y and R alleles always stay together, generation after generation? Or are seed color and seed shape inherited independently? **Figure 14.8** shows how a **dihybrid cross**, a cross between F1 dihybrids, can determine which of these two hypotheses is correct.

Figure 14.8

Inquiry: Do the alleles for one character segregate into gametes dependently or independently of the alleles for a different character?

Experiment

To follow the characters of seed color and seed shape through the F2 generation, Mendel crossed a true-breeding plant with yellow round seeds with a true-breeding plant with green wrinkled seeds, producing dihybrid F1 plants. Self-pollination of the F1 dihybrids produced the F2 generation. The two hypotheses (dependent and independent "assortment" of the two genes) predict different phenotypic ratios.

Two characteristics (seed shape and seed color) traced from parental generation to F 2 generation to test whether assortment is dependent or independent. Resulting phenotypic ratio of 9 to 3 to 3 to 1 supports independent assortment.

See table next tab

P generation: Round yellow seed (Y Y R R, all uppercase) crossed with wrinkled green seed (y y r r, all lowercase). Gametes from Y Y R R, all uppercase = 100% Y R, both uppercase and gametes from y y r r, all lowercase equals to 100% y r, both lowercase.

F1 generation equals to All round yellow seeds (all with genotypes uppercase Y lowercase y uppercase R lowercase r).

Hypothesis of dependent assortment predicts a phenotypic ratio of 3 round, yellow to 1 wrinkled, green. See table 14.8a.

Hypothesis of independent assortment predicts nine sixteenths round yellow seeds, three sixteenths round green seeds, three sixteenths wrinkled yellow seeds and one sixteenth wrinkled green seeds; or a phenotypic ratio of 9 to 3 to 3 to 1.

Gametes	one-fourth Y R, both uppercase	one-fourth uppercase Y lowercase r	uppercase R one-fourth lowercase y	one-fourth y r, both lowercase
one-fourth Y R, both uppercase	Y Y R R, all uppercase. Phenotype: Round, yellow	Y Y, both uppercase, uppercase R lowercase r. Phenotype: Round, yellow	Uppercase Y lowercase y, RR both uppercase. Phenotype: Round, yellow	Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, yellow
one-fourth uppercase Y lowercase r	Y Y, both uppercase, uppercase R lowercase r.	Y Y, both uppercase, rr both lowercase. Phenotype: wrinkled, yellow	Uppercase Y lowercase y Uppercase R lowercase r.	Uppercase Y lowercase y, rr, both lowercase.

	Phenotype: Round, yellow		Phenotype: Round, yellow	Phenotype: wrinkled, yellow
one-fourth lowercase y uppercase R	Uppercase Y lowercase y RR both uppercase. Phenotype: Round, yellow	Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, yellow	y y, both lowercase, RR both uppercase. Phenotype: Round, green.	y y, both lowercase, uppercase R lowercase r. Phenotype: Round, green.
one-fourth yr both lowercase	Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, yellow	Uppercase Y lowercase y, r r, both lowercase. Phenotype: wrinkled, yellow	y y, both lowercase, uppercase R lowercase r. Phenotype: Round, green.	y y r r, all lowercase. Phenotype: wrinkled, green.

Results

Results: 315 round yellow seeds, 108 round green seeds, 101 wrinkled yellow seeds and 32 wrinkled green seeds. Experimental phenotypic ratio = approximately 9 to 3 to 3 to 1.

Gametes	one-half Y R, both uppercase	one-half y r, both lowercase
one half Y R, both uppercase	Y Y R R, all uppercase Phenotype: round, yellow	Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, Yellow
one half y r, both lowercase	Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: round, yellow	y y r r all lowercase Phenotype: wrinkled, green

Conclusion

The results support the hypothesis of independent assortment, the only one that predicts two newly observed phenotypes: green round seeds and yellow wrinkled seeds (see the right-hand Punnett square). The alleles for each gene segregate independently of those of the other, and the two genes are said to assort independently.

Data from G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brünn*4:3–47 (1866).

WHAT IF? *Suppose Mendel had transferred pollen from an* F_1 *plant to the carpel of a plant that was homozygous recessive for both genes. Set up the cross and draw Punnett squares that predict the offspring for both hypotheses. Would this cross have supported the hypothesis of independent assortment equally well?*

*For suggested answer, see*Appendix A.*

Animation

Mendel's Cross of Two Characters: Seed Shape and Seed Color

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Animation

A Simplified Cross of Two Characters in Humans

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Animation

Crosses of Two Characters in "MendAliens"

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The F_1 plants, of genotype YyRr, exhibit both dominant phenotypes, yellow seeds with round shapes, no matter which hypothesis is correct. The key step in the experiment is to see what happens when F_1 plants self-pollinate and produce F_2 offspring. If the hybrids must transmit their alleles in the same combinations in which the alleles were inherited from the P generation, then the F_1 hybrids will produce only two classes

of gametes: YR and yr. As shown on the left side of Figure 14.8, this “dependent assortment” hypothesis predicts that the phenotypic ratio of the F2 generation will be 3:1, just as in a monohybrid cross:

Three-fourths yellow round, and one-fourth green wrinkled.

The alternative hypothesis is that the two pairs of alleles segregate independently of each other. In other words, genes are packaged into gametes in all possible allelic combinations, as long as each gamete has one allele for each gene (see Figure 13.11). In our example, an F1 plant will produce four classes of gametes in equal quantities: YR,Yr,yR, and yr. If sperm of the four classes fertilize eggs of the four classes, there will be 16(4×4) equally probable ways in which the alleles can combine in the F2 generation, as shown on the right side of Figure 14.8. These combinations result in four phenotypic categories with a ratio of 9:3:3:1 (9 yellow round to 3 green round to 3 yellow wrinkled to 1 green wrinkled):

Nine-sixteenths yellow ground to three-sixteenths green round to three-sixteenths yellow wrinkled to one-sixteenth green wrinkled.

When Mendel did the experiment and classified the F2 offspring, his results were close to the predicted 9:3:3:1 phenotypic ratio, supporting the hypothesis that the alleles for one gene—the gene for seed color, for example—segregate into gametes independently of the alleles of any other gene, such as the gene for seed shape.

Mendel tested his seven pea characters in various dihybrid combinations and always observed a 9:3:3:1 phenotypic ratio in the F2 generation. Is this consistent with the 3:1 phenotypic ratio seen for the monohybrid cross shown in Figure 14.5? To answer this question, count the number of yellow and green peas, ignoring shape, and calculate the ratio. The results of Mendel’s dihybrid experiments are the basis for what we now call the **law of independent assortment**, which states that *two or more genes assort independently—that is, each pair of alleles segregates independently of any other pair of alleles—during gamete formation.*

This law applies only to genes (allele pairs) located on different chromosomes (that is, on chromosomes that are not homologous) or, alternatively, to genes that are very far apart on the same chromosome. (This will be explained in Concept 15.3, along with the more complex inheritance patterns of genes located near each other, alleles of which tend to be inherited together.) All the pea characters Mendel chose for analysis were controlled by genes on different chromosomes or were far apart on the same chromosome; this situation greatly simplified interpretation of his multicharacter pea crosses. All the examples we consider in the rest of this chapter involve genes located on different chromosomes.

Animation

Independent Assortment

image

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As you learned in Figure 13.6b, meiosis in plants produces spores, not gametes. In flowering plants like the pea, each spore develops into a microscopic haploid gametophyte that contains only a few cells and is located on the parent plant. The gametophyte produces sperm, in pollen grains, and eggs, in the carpel. For simplicity, we will not include the gametophyte stage in our discussion of fertilization in plants.